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Towards a Formal Theory of Representational Compositionality

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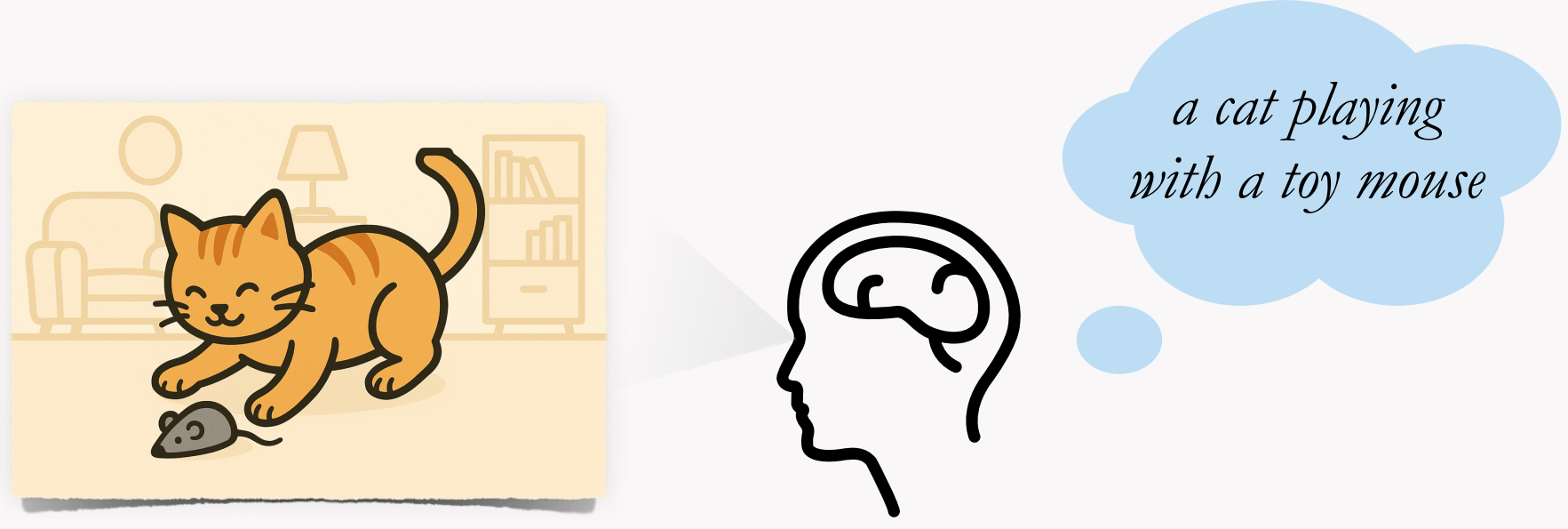
Code

Paper

Motivation

Compositionality is essential to **intelligence**


a cat playing
with a toy mouse

Question

What is compositionality, really?
How can we **define it formally**?

Colloquially
"The meaning of an expression is determined by its structure and the meanings of its parts"

In a neural representation, what do these things mean???

Our definition

Compositional representations are both **expressive** and describable as a **simple function of** symbolic **parts**

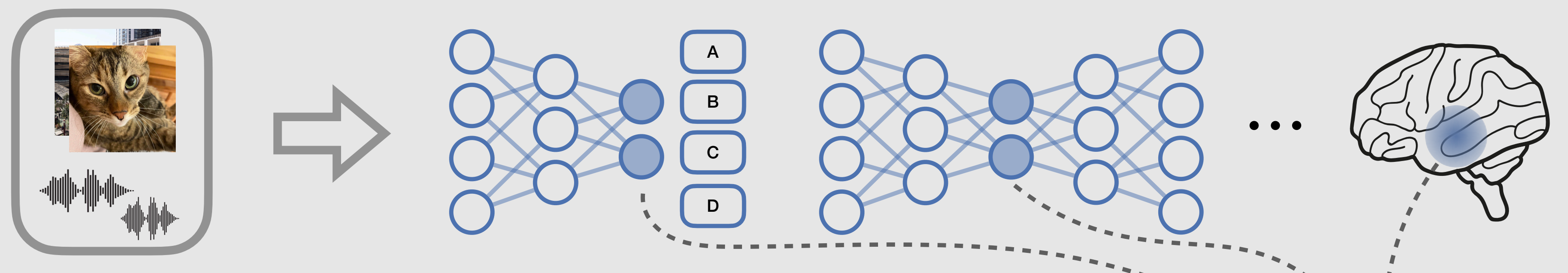
Simple?
Expressive?
Describable?

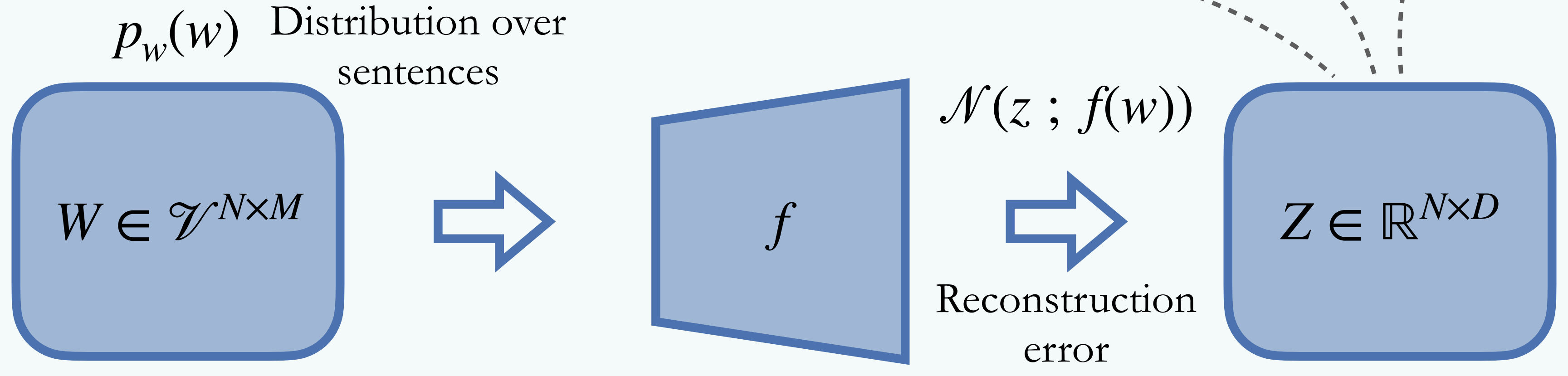
Grounded in **algorithmic information theory**

Representational Compositionality

Compress the representation as a **function of parts**

Representation samples from a pre-trained model, brain, etc.




$p_w(w)$ Distribution over sentences
$W \in \mathcal{V}^{N \times M}$
f
$\mathcal{N}(z\ ;\ f(w))$
Reconstruction error
$Z \in \mathbb{R}^{N \times D}$

Sentences
Discrete symbol sequences describing the representation

Semantics
Mapping from sentences to the representation

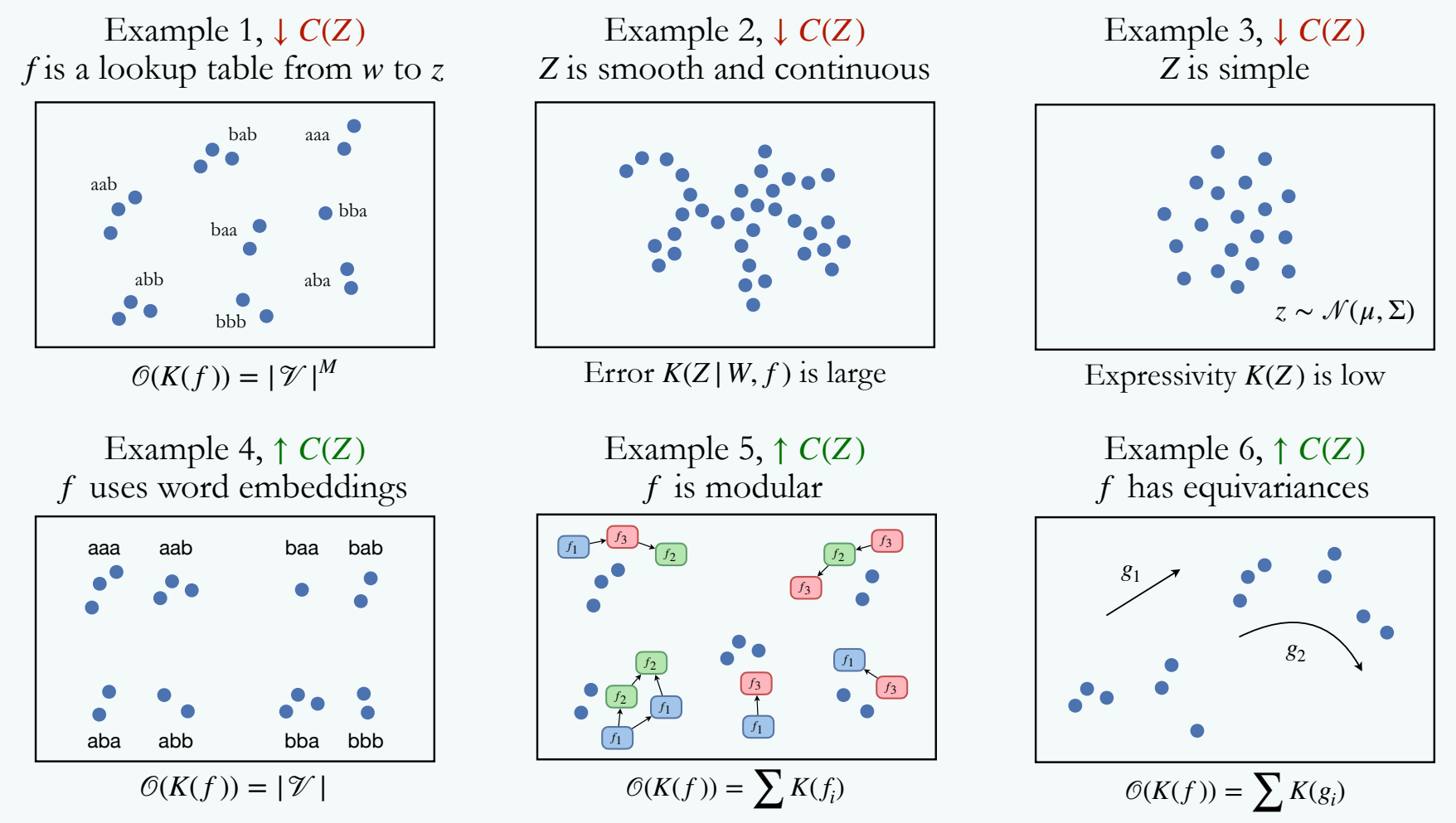
Representation
Latent model representation

Kolmogorov complexity
Optimal compression: length of shortest program outputting object

$$K(Z) \approx \min_{p_w, W, f} K(p_w) + K(W \mid p_w) + K(f) + K(Z \mid W, f)$$

Representational compositionality

$$C(Z) = \frac{\text{Expressivity of the representation}}{\text{Systematicity of the semantics/rules}} = \frac{K(Z)}{K(Z \mid W)} = \frac{K(Z)}{K(f) + K(Z \mid W, f)}$$


Example 1, ↓ $C(Z)$
f is a lookup table from w to z
aab
bab
aaa
baa
bba
abb
aba
bbb
$\mathcal{O}(K(f)) = |\mathcal{V}|^M$
Example 2, ↓ $C(Z)$
Z is smooth and continuous
Error $K(Z \mid W, f)$ is large
Example 3, ↓ $C(Z)$
Z is simple
$z \sim \mathcal{N}(\mu, \Sigma)$
Expressivity $K(Z)$ is low
Example 4, ↑ $C(Z)$
f uses word embeddings
aaa
aab
baa
bab
aba
abb
bba
bbb
$\mathcal{O}(K(f)) = |\mathcal{V}|$
Example 5, ↑ $C(Z)$
f is modular
f_1
f_2
f_3
$\mathcal{O}(K(f)) = \sum K(f_i)$
Example 6, ↑ $C(Z)$
f has equivariances
g_1
g_2
$\mathcal{O}(K(f)) = \sum K(g_i)$

Language system compositionality

W given

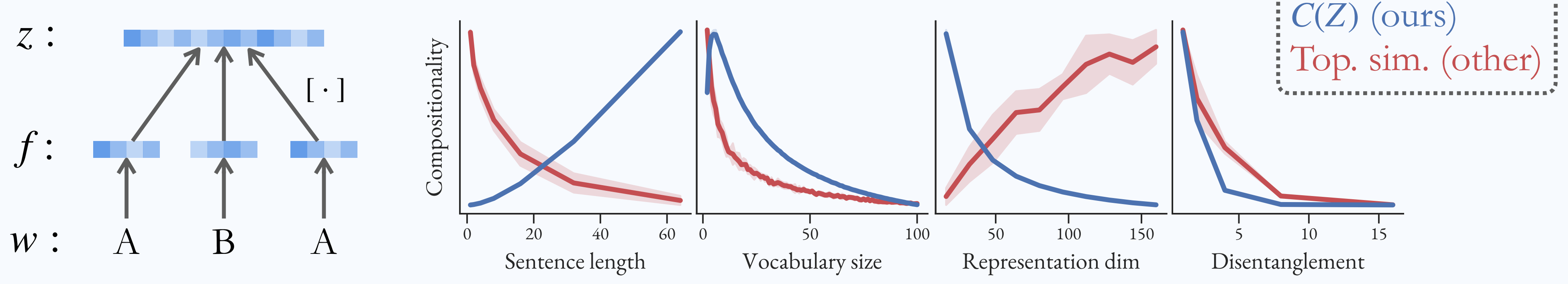
$$C^L(Z) = \frac{K(Z)}{K(Z \mid W^L)} = \frac{K(Z)}{K(f^L) + K(Z \mid W^L, f^L)}$$

Expressive $\iff \uparrow K(Z)$
Simple semantics $\iff \downarrow K(f)$
Describable from parts $\iff \downarrow K(Z \mid W, f)$

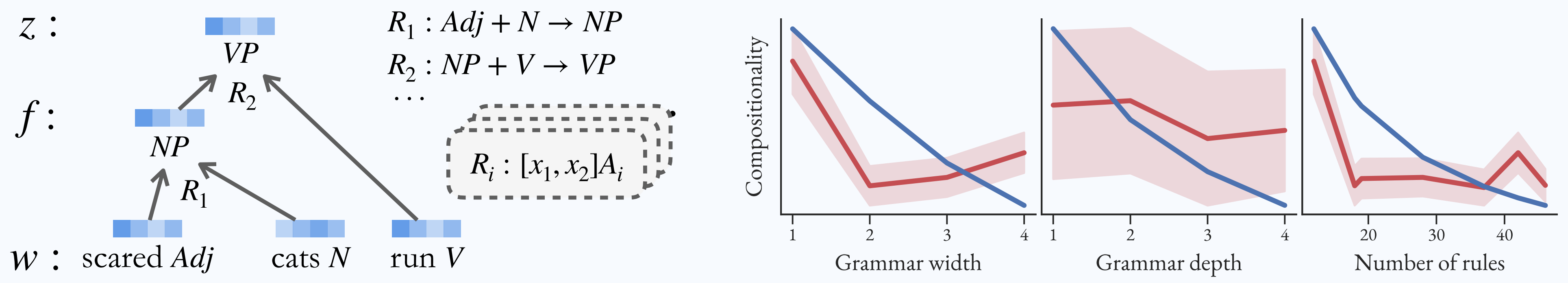
Results

Does $C(Z)$ match intuitions on **synthetic** data?

Lookup table representations



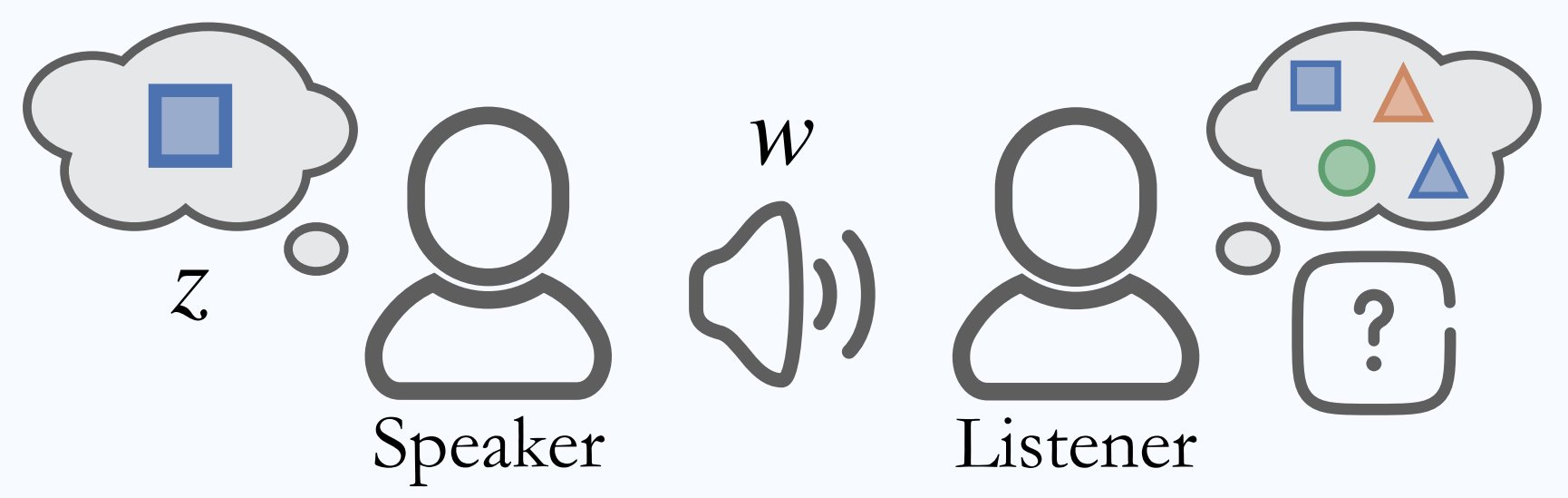
Synthetic grammar representations



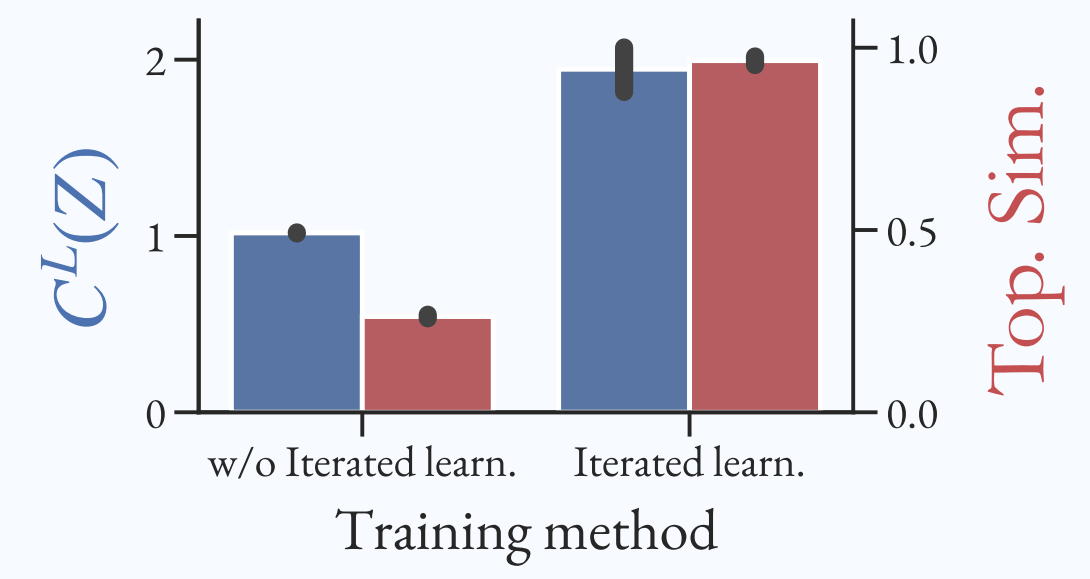
$C(Z)$ matches intuitions for both synthetic datasets
Topological similarity (another metric) does not

What does $C^L(Z)$ have to say for **real** data?

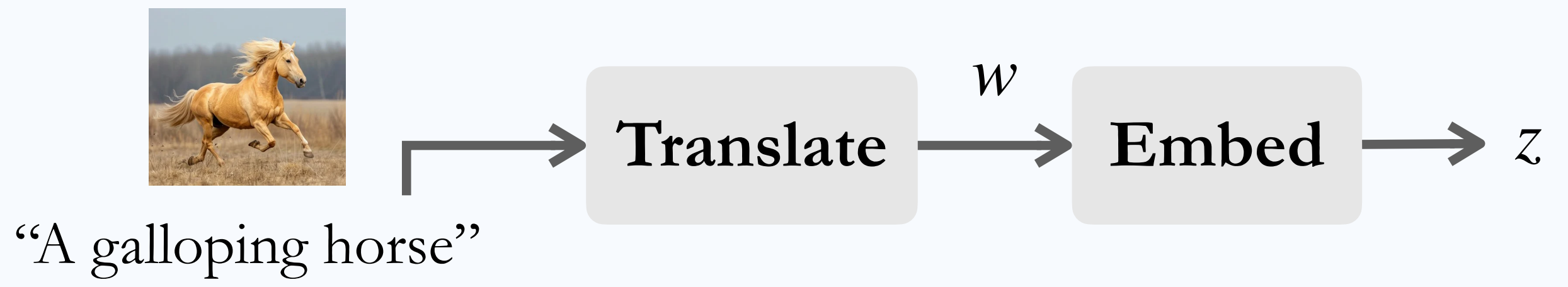
Emergent languages from multi-agent communication games

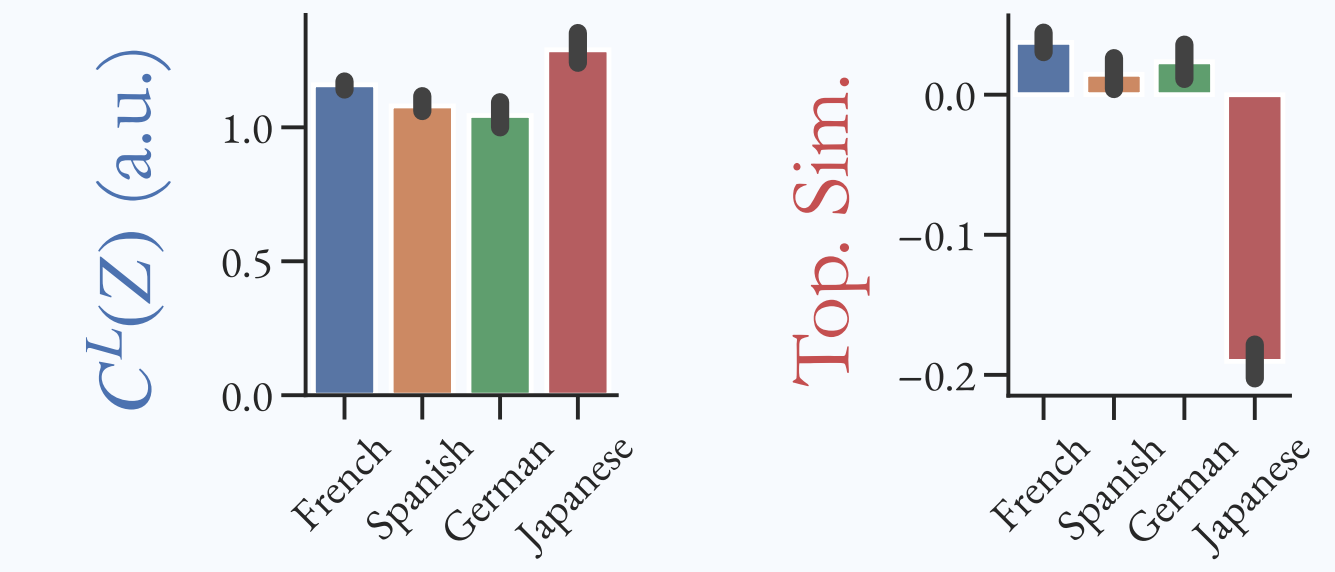

z
Speaker
w
Listener
?

"Iterated learning"
↓
More compositional languages



Natural language with representations from LLM sentence embeddings


"A galloping horse"
Translate
w
Embed
z



$C^L(Z)$ reproduces past results on emergent languages and says **natural languages are equally compositional**

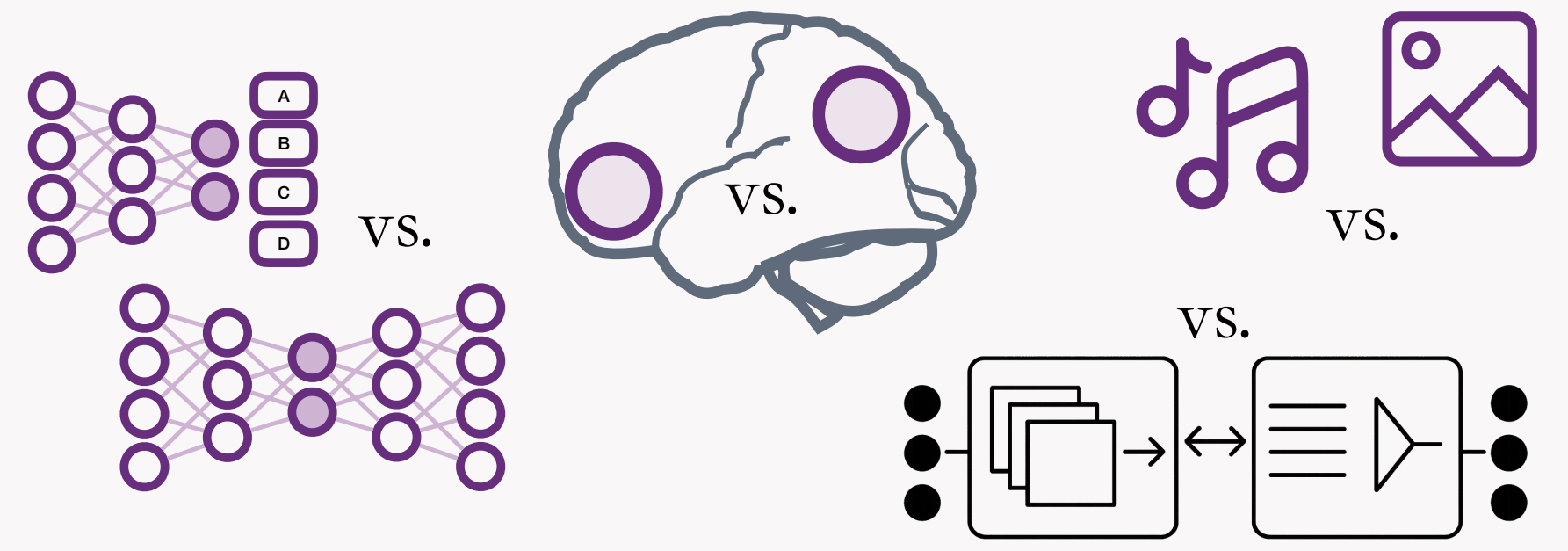
Conclusion

Our definition **accounts for** and **extends** intuitions of compositionality

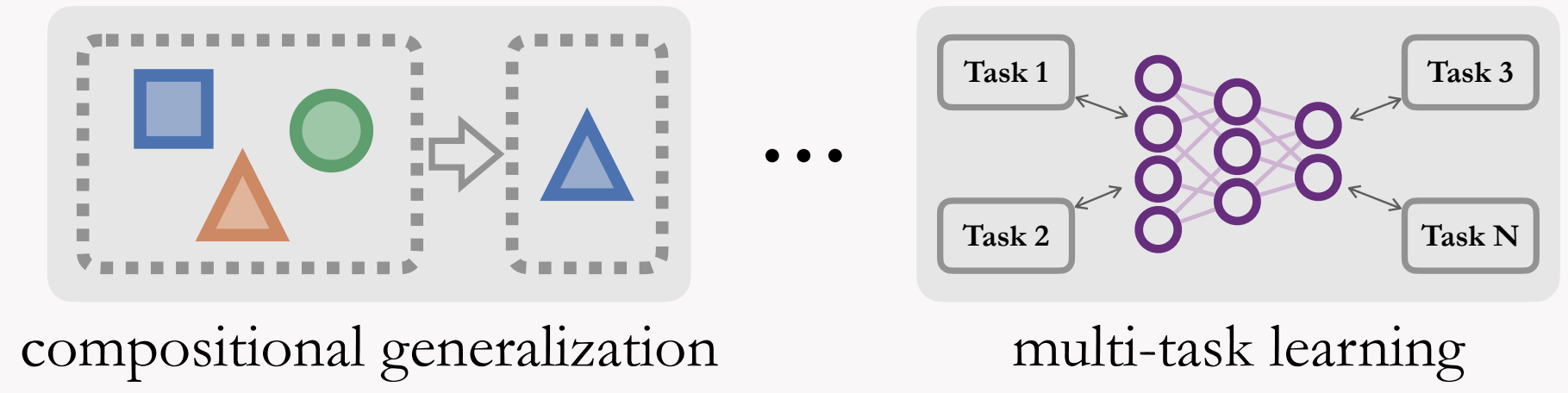
Future work

Estimate compositionality of *representations* $C(Z)$ in real data (not only *language systems* $C^L(Z)$)

Compare representations across…


A
B
C
D
vs.
vs.
vs.
vs.

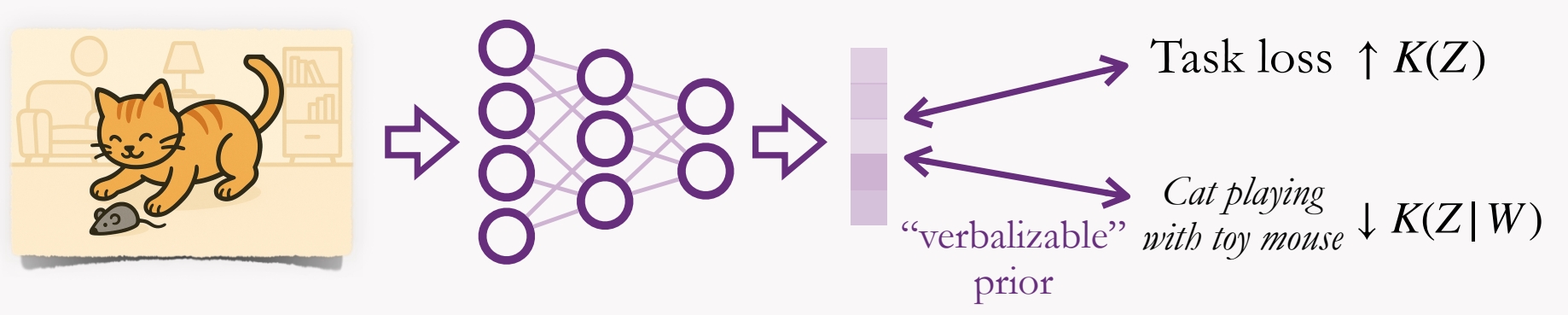
Investigate relationship to…


…
Task 1
Task 2
Task 3
Task N

compositional generalization

multi-task learning

Design soft inductive biases


Task loss ↑ $K(Z)$
Cat playing with toy mouse ↓ $K(Z \mid W)$
"verbalizable" prior

Limitations

1. $C(Z)$, like Kolmogorov complexity, is uncomputable (but can be estimated)
2. Did not compare to related reconstruction-based definitions
3. Only defined w.r.t. discrete parts